

Hints & Solutions

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Q1 (4) - Single Correct

$$\frac{\sin \theta}{\cos 3\theta} = \frac{1}{2} \frac{2 \sin \theta \cos \theta}{\cos 3\theta \cos \theta} = \frac{1}{2} \frac{\sin 2\theta}{\cos 3\theta \cos \theta}$$

$$\text{Similarly } \frac{\sin 3\theta}{\cos 9\theta} = \frac{1}{2} (\tan 9\theta - \tan 3\theta)$$

$$\text{and } \frac{\sin 9\theta}{\cos 27\theta} = \frac{1}{2} (\tan 27\theta - \tan 9\theta)$$

$$\text{so } \frac{1}{2} (\tan x - \tan \theta) = \frac{1}{2} (\tan 27\theta - \tan \theta)$$

$$\Rightarrow x = 27\theta$$

Q2 (2) - Single Correct

$$4x^2 \pm 4x \sin \theta - (1 - \sin^2 \theta)$$

$$= 4x^2 \pm 4x \sin \theta + \sin^2 \theta - 1$$

$$= (2x \pm \sin \theta)^2 - 1 \geq -1$$

Hence the required value is -1 .

Q3 (3) - Single Correct

We can write the given expression as

$$\cos^4\left(\frac{\pi}{8}\right) + \cos^4\left(\frac{3\pi}{8}\right) + \cos^4\left(\pi - \frac{3\pi}{8}\right) + \cos^4\left(\pi - \frac{\pi}{8}\right)$$

$$= 2 \left[\cos^4\left(\frac{\pi}{8}\right) + \cos^4\left(\frac{3\pi}{8}\right) \right]$$

$$= 2 \left[\cos^4\left(\frac{\pi}{8}\right) + \sin^4\left(\frac{\pi}{2} - \frac{3\pi}{8}\right) \right]$$

$$= 2 \left[\cos^4\left(\frac{\pi}{8}\right) + \sin^4\left(\frac{\pi}{8}\right) \right]$$

$$= 2 \left[\left(\cos^2\left(\frac{\pi}{8}\right) + \sin^2\left(\frac{\pi}{8}\right) \right)^2 - 2 \cos^2\left(\frac{\pi}{8}\right) \sin^2\left(\frac{\pi}{8}\right) \right]$$

$$= 2 \left[1 - \frac{1}{2} \sin^2\left(\frac{\pi}{4}\right) \right] = 2 \left[1 - \frac{1}{2} \times \frac{1}{2} \right] = \frac{3}{2}$$

Q4 (4) - Single Correct

The given expression is equal to

$$(\sin 47^\circ + \sin 61^\circ) - (\sin 11^\circ + \sin 25^\circ)$$

$$= 2 \sin 54^\circ \cos 7^\circ - 2 \sin 18^\circ \cos 7^\circ$$

$$= 2 \cos 7^\circ (\sin 54^\circ - \sin 18^\circ)$$

$$= 2 \cos 7^\circ \left[\frac{\sqrt{5} + 1}{4} - \frac{\sqrt{5} - 1}{4} \right] = \cos 7^\circ.$$

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Q5 (4) - Single Correct

From the given equation we get $S_1 = \tan \theta_1 + \tan \theta_2 + \tan \theta_3 + \tan \theta_4 = \sin 2\beta$

$$S_2 = \Sigma \tan \theta_1 \tan \theta_2 = \cos 2\beta$$

$$S_3 = \Sigma \tan \theta_1 \tan \theta_2 \tan \theta_3 = \cos \beta$$

$$\text{and } S_4 = \tan \theta_1 \tan \theta_2 \tan \theta_3 \tan \theta_4 = -\sin \beta$$

$$\begin{aligned} \text{Now } \tan(\theta_1 + \theta_2 + \theta_3 + \theta_4) &= \frac{S_1 - S_3}{1 - S_2 + S_4} \\ &= \frac{\sin 2\beta - \cos \beta}{1 - \cos 2\beta - \sin \beta} = \frac{\cos \beta (2 \sin \beta - 1)}{\sin \beta (2 \sin \beta - 1)} = \cot \beta. \end{aligned}$$

Q6 (3) - Single Correct

$$\text{Let } \theta_k = \frac{\pi}{4} + \frac{k\pi}{6}, 0 \leq k \leq 13$$

$$\text{Note that } \theta_k - \theta_{k-1} = \frac{\pi}{6}$$

$$\text{Let } S = \sum_{k=1}^{13} \frac{1}{\sin \theta_{k-1} \sin \theta_k}$$

$$= 2 \sum_{k=1}^{13} \frac{\sin \frac{\pi}{6}}{\sin \theta_{k-1} \sin \theta_k}$$

$$= 2 \sum_{k=1}^{13} \frac{\sin(\theta_k - \theta_{k-1})}{\sin \theta_{k-1} \sin \theta_k}$$

$$= 2 \sum_{k=1}^{13} (\cot \theta_{k-1} - \cot \theta_k)$$

$$= 2 (\cot \theta_0 - \cot \theta_{13})$$

$$= 2 \left[1 - \cot \left(\frac{\pi}{4} + \frac{13\pi}{6} \right) \right]$$

$$= 2 \left[1 - \frac{\cot \frac{\pi}{4} \cos \frac{\pi}{6} - 1}{\cot \frac{\pi}{4} + \cot \frac{\pi}{6}} \right]$$

$$= 2 \left[1 - \frac{\sqrt{3}-1}{\sqrt{3}+1} \right]$$

$$= 2(\sqrt{3}-1)$$

Q7 (1) - Single Correct

From the given relations we have

$$\prod_{i=1}^n (\cos \alpha_i) = \prod_{i=1}^n (\sin \alpha_i)$$

$$\Rightarrow \prod_{i=1}^n (\cos^2 \alpha_i) = \prod_{i=1}^n (\cos \alpha_i \sin \alpha_i) = \prod_{i=1}^n \left(\frac{\sin 2\alpha_i}{2} \right)$$

$$\text{Since } 0 \leq \alpha_i \leq \pi/2 \Rightarrow 0 \leq 2\alpha_i \leq \pi$$

$$\therefore \prod_{i=1}^n (\cos^2 \alpha_i) \leq \frac{1}{2^n} \text{ as max. value of } \sin 2\alpha_i \text{ is } 1$$

for all i .

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$$\Rightarrow \prod_{i=1}^n (\cos \alpha_i) \leq \frac{1}{2^{n/2}}$$

So the maximum value of the given expression is $\frac{1}{2^{n/2}}$.

Q8 (3) - Single Correct

Let $x_0 = \cos \theta$, then $x_1 = \sqrt{\frac{1}{2}(1 + \cos \theta)}$
 $= \cos \theta/2$, $x_2 = \cos(\theta/2^2)$, $x_3 = \cos(\theta/2^3)$, $\dots$ and so on.

so that $\left[\frac{\sqrt{x_1 x_2 x_3 \dots \text{to infinite}}}{\cos \frac{\theta}{2} \cos \frac{\theta}{2^2} \dots \cos \frac{\theta}{2^n} \dots \text{infinite}} \right]$

$$= \frac{\sin \theta}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}$$

$$= \frac{\sin \theta}{2^2 \sin \frac{\theta}{2^2} \cos \frac{\theta}{2^2}}$$

$$= \frac{\sin \theta}{2^n \sin \frac{\theta}{2^n} \cos \frac{\theta}{2^n}}$$

$$= \lim_{n \rightarrow \infty} \theta \left(\frac{\sin \frac{\theta}{2^n}}{\frac{\theta}{2^n}} \right) \frac{1}{\cos \frac{\theta}{2^{n+1}}} = \theta$$

$$\text{so that } \cos \left[\frac{\sqrt{1-x_0^2}}{x_1 x_2 \dots \text{inf.}} \right] = \cos \theta = x_0$$

Q9 (3) - Single Correct

$$\sqrt{3} \sec x + \operatorname{cosec} x + 2(\tan x - \cot x) = 0$$

$$\Rightarrow \frac{\sqrt{3}}{2} \frac{1}{\cos x} + \frac{1}{2} \frac{1}{\sin x} + \frac{\sin x}{\cos x} - \frac{\cos x}{\sin x} = 0$$

$$\Rightarrow \frac{1}{2} \cos x + \frac{\sqrt{3}}{2} \sin x = \cos^2 x - \sin^2 x$$

$$\Rightarrow \cos \left(x - \frac{\pi}{3} \right) = \cos 2x$$

$$\Rightarrow 2x = 2n\pi \pm \left(x - \frac{\pi}{3} \right), n \in \mathbf{I}$$

Taking positive sign, we get,

$$x = 2n\pi - \frac{\pi}{3}, n \in \mathbf{I}$$

$$\text{As } x \in (-\pi, \pi), x = -\frac{\pi}{3}$$

Taking negative sign, we get

$$2x = 2n\pi - \left(x - \frac{\pi}{3} \right), n \in \mathbf{I}$$

$$\Rightarrow x = \frac{2}{3}n\pi + \frac{1}{9}\pi, n \in \mathbf{I}$$

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As $x \in (-\pi, \pi)$, we get

$$x = -\frac{2}{3}\pi + \frac{1}{9}\pi, \frac{1}{9}\pi, \frac{2}{3}\pi + \frac{1}{9}\pi$$

$\therefore$ Sum of solutions is

$$-\frac{\pi}{3} + \left(-\frac{2}{3}\pi + \frac{1}{9}\pi\right) + \frac{1}{9}\pi + \left(\frac{2}{3}\pi + \frac{1}{9}\pi\right) = 0.$$

Q10 (4) - Single Correct

$$\sin x + 2 \sin 2x - \sin 3x = 3 \quad (1)$$

Note that $x = \pi/2$ does not satisfy (1). Therefore, we may assume that $0 < x < \pi, x \neq \pi/2$. We write (1) as

$$\sin x + 4 \sin x \cos x - (3 \sin x - 4 \sin^3 x) = 3$$

$$\Rightarrow 4 \sin x \cos x + 4 \sin^3 x - 2 \sin x = 3$$

$$\Rightarrow 4 \cos x + 4(1 - \cos^2 x) - 2 = \frac{3}{\sin x}$$

$$\Rightarrow 4 \cos^2 x - 4 \cos x + 1 = 3 \left(1 - \frac{1}{\sin x}\right)$$

$$\Rightarrow (2 \cos x - 1)^2 = 3 \left(1 - \frac{1}{\sin x}\right) \quad (2)$$

As $0 < x < \pi, x \neq \pi/2, 0 < \sin x < 1$

$$\Rightarrow 1 - \frac{1}{\sin x} < 0$$

$\therefore$ RHS of (2) < 0

and LHS of (2) ≥ 0 . This not possible.

Thus, (1) has no solution in $(0, \pi)$

Q11 (3) - Single Correct

$$2 \sin^2 \theta - \cos 2\theta = 0$$

$$\Rightarrow 2 \sin^2 \theta - (1 - 2 \sin^2 \theta) = 0$$

$$\Rightarrow 4 \sin^2 \theta = 1 \Rightarrow \sin \theta = \pm 1/2$$

$$\text{and } 2 \cos^2 \theta - 3 \sin \theta = 0$$

$$\Rightarrow 2(1 - \sin^2 \theta) - 3 \sin \theta = 0 \Rightarrow 2(1 - \sin^2 \theta) - 3 \sin \theta = 0 \Rightarrow 2 \sin^2 \theta + 3 \sin \theta - 2 = 0$$

$$\Rightarrow \sin \theta = \frac{-3 \pm \sqrt{9+16}}{4} = \frac{-3 \pm 5}{4}$$

$$\Rightarrow \sin \theta = 1/2 \text{ as } -1 \leq \sin \theta \leq 1$$

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So $\sin \theta = 1/2$ satisfies both the equations

$$\Rightarrow \theta = \frac{\pi}{6}, - \text{ as } \theta \in [0, 2\pi]$$

giving two values of θ .

Q12 (1) - Single Correct

$\cos(\pi\sqrt{x-4})$ is defined if $x \geq 4$

$$\cos(\pi\sqrt{x}) \cos(\pi\sqrt{x-4}) = 1$$

$$\Rightarrow \cos(\pi\sqrt{x}) = 1, \cos(\pi\sqrt{x-4}) = 1$$

$$\Rightarrow \pi\sqrt{x} = 2m\pi, \pi\sqrt{x-4} = 2n\pi (m, n \in \mathbb{I})$$

$$\Rightarrow x = 4m^2, x-4 = 4n^2$$

$$\Rightarrow m^2 - n^2 = 1 \Rightarrow m = 1, n = 0 \Rightarrow x = 4$$

So $x = 4$ is the only solution of the equation

Q13 (3) - Single Correct

$$|\sin x| \geq 0 \Rightarrow \frac{|\sin x|}{1+|\sin x|} < 1$$

$$\Rightarrow 1 - \frac{|\sin x|}{1+|\sin x|} > 0$$

So the given equation becomes

$$1 - \frac{|\sin x|}{1+|\sin x|} \geq \frac{2}{3}$$

$$\Rightarrow \frac{1}{3} \geq \frac{|\sin x|}{1+|\sin x|} \Rightarrow |\sin x| \leq \frac{1}{2}$$

$$\Rightarrow -\frac{1}{2} \leq \sin x \leq \frac{1}{2} \Rightarrow \sin x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

Q14 (2) - Single Correct

Let $\cos^{-1} x = y$.

Then $x = \cos y$, so that $1/2 \leq x \leq 1$, or $0 \leq y \leq \pi/3$, and

$$\begin{aligned} \frac{x}{2} + \frac{1}{2}\sqrt{3-3x^2} &= \frac{1}{2}\cos y + \frac{\sqrt{3}}{2}\sin y \\ &= \cos \frac{\pi}{3} \cos y + \sin \frac{\pi}{3} \sin y = \cos\left(\frac{\pi}{3} - y\right) \end{aligned}$$

$$\Rightarrow \cos^{-1}\left(\frac{x}{2} + \frac{1}{2}\sqrt{3-3x^2}\right) = \frac{\pi}{3} - y$$

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because $\cos^{-1}(\cos x) = x$ if $0 \leq x \leq \pi$.

Therefore, the given expression is equal to $y + \pi/3 - y = \pi/3$.

Q15 (2) - Single Correct

The n th term of the series can be written as

$$\begin{aligned} u_n &= \cot^{-1}(2n^2) = \tan^{-1}(1/2n^2) \\ &= \tan^{-1}\left(\frac{(2n+1) - (2n-1)}{1 + (2n+1)(2n-1)}\right) \\ &= \tan^{-1}(2n+1) - \tan^{-1}(2n-1) \end{aligned}$$

so that $u_1 = \tan^{-1} 3 - \tan^{-1} 1$

$$u_2 = \tan^{-1} 5 - \tan^{-1} 3$$

$$u_3 = \tan^{-1} 7 - \tan^{-1} 5$$

and the sum to n terms is therefore written as

$$\begin{aligned} S_n &= \tan^{-1}(2n+1) - \tan^{-1} 1 \\ \Rightarrow \lim_{n \rightarrow \infty} S_n &= \pi/2 - \pi/4 = \pi/4 \end{aligned}$$

Q16 (4) - Single Correct

$$\text{We have } b \sin^{-1} x + b \cos^{-1} x = \frac{b\pi}{2}$$

$$\text{and } a \sin^{-1} x - b \cos^{-1} x = c \quad (\text{given})$$

$$\Rightarrow (a+b) \sin^{-1} x = \frac{b\pi}{2} + c$$

$$\Rightarrow \sin^{-1} x = \frac{(b\pi)/2 + c}{a+b}$$

$$\text{Similarly } \cos^{-1} x = \frac{(a\pi)/2 - c}{a+b}$$

$$\text{so that } a \sin^{-1} x + b \cos^{-1} x = \frac{\pi ab + c(a-b)}{a+b}$$

Q17 (4) - Single Correct

If we denote $\cos^{-1} x$ by y , then since $0 \leq \cos^{-1} x \leq \pi \Rightarrow 0 \leq 2y \leq 2\pi$

$$\text{Also since } -\pi/2 \leq \sin^{-1}(2x\sqrt{1-x^2}) \leq \pi/2 \Rightarrow -\pi/2 \leq$$

$$\sin^{-1} \sin(2y) \leq \pi/2$$

$$\Rightarrow -\pi/2 \leq 2y \leq \pi/2$$

$$\text{From (1) and (2) we find } 0 \leq 2y \leq \pi/2 \Rightarrow 0 \leq y \leq \pi/4 \Rightarrow 0 \leq \cos^{-1} x \leq \pi/4$$

which holds if $1/\sqrt{2} \leq x \leq 1$.

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Q18 (1,2,3,4) - Multiple Correct

We have $A = \tan^{-1}(1/7)$, or $\tan A = 1/7$.

$$\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A} = \frac{1 - 1/49}{1 + 1/49} = \frac{48}{50} = \frac{24}{25}$$

Also, from $B = \tan^{-1}(1/3)$, or $\tan B = 1/3$, we get $\sin 2B = \frac{2 \tan B}{1 + \tan^2 B} = \frac{2(1/3)}{1 + (1/9)} = \frac{2}{3} \cdot \frac{9}{10} = \frac{3}{5}$

$$\text{and } \cos 2B = \frac{1 - 1/9}{1 + 1/9} = \frac{8}{10} = \frac{4}{5}$$

$$\therefore \sin 4B = 2 \sin 2B \cos 2B = 2 \cdot \frac{3}{5} \cdot \frac{4}{5} = \frac{24}{25}$$

Q19 (1) - Multiple Correct

The given equation can be written as $\sin^{-1} 6x = -\pi/2 - \sin^{-1} 6\sqrt{3}x$

$$\Rightarrow \sin(\sin^{-1} 6x) = \sin(-\pi/2 - \sin^{-1} 6\sqrt{3}x)$$

$$\Rightarrow 6x = -\cos(\sin^{-1} 6\sqrt{3}x)$$

$$\Rightarrow 6x = -\sqrt{1 - 108x^2}$$

Squaring both sides, we get

$$36x^2 = 1 - 108x^2 \Rightarrow 144x^2 = 1$$

$$\Rightarrow x = \pm 1/12$$

Substituting $x = -1/12$ in the given equation, we get

$$\text{LHS} = \sin^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{6} - \frac{\pi}{3} = -\frac{\pi}{2} = \text{RHS}$$

Therefore, $x = -1/12$ is a root of the given equation.

Again, substituting $x = 1/12$ in the given equation, we get $\text{LHS} = \pi/2$,

so that $x = 1/12$ is an extraneous root of the given equation. Hence $x = -1/12$ is the only root.

Q20 (1,2) - Multiple Correct

$$\theta = \tan^{-1}(2 \tan^2 \theta) - \tan^{-1}((1/3) \tan \theta)$$

$$\Rightarrow \tan \theta = \frac{2 \tan^2 \theta - (1/3) \tan \theta}{1 + (2/3) \tan^3 \theta}$$

$$\Rightarrow \tan \theta \frac{2 \tan \theta - (1/3)}{1 + (2/3) \tan^3 \theta} - 1 = 0$$

which is true if $\tan \theta = 0$

$$\text{or } \frac{2 \tan \theta - (1/3)}{1 + (2/3) \tan^3 \theta} = 1 \Rightarrow \tan^3 \theta - 3 \tan \theta + 2 = 0$$

$$\Rightarrow (\tan \theta - 1)^2 (\tan \theta + 2) = 0 \text{ which holds if}$$

$$\tan \theta = 1 \text{ or } \tan \theta = -2$$

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Q21 (3) - Multiple Correct

We have

$$S_1 = \sum x_1 = \sin 2\beta$$

$$S_2 = \sum x_1 x_2 = \cos 2\beta$$

$$S_3 = \sum x_1 x_2 x_3 = \cos \beta$$

$$S_4 = x_1 x_2 x_3 x_4 = -\sin \beta$$

$$\text{so that } \sum_{i=1}^4 \tan^{-1} x_i = \tan^{-1} \frac{S_1 - S_3}{1 - S_2 + S_4}$$

$$= \tan^{-1} \frac{\sin 2\beta - \cos \beta}{1 - \cos 2\beta - \sin \beta}$$

$$= \tan^{-1} \frac{\cos \beta (2 \sin \beta - 1)}{\sin \beta (2 \sin \beta - 1)}$$

$$= \tan^{-1} \cot \beta = \tan^{-1} (\tan(\pi/2 - \beta))$$

$$= \pi/2 - \beta$$

Q22 (2,3) - Multiple Correct

$$(1 - \tan \theta)(1 + \tan \theta) \sec^2 \theta + 2^{\tan^2 \theta} = 0$$

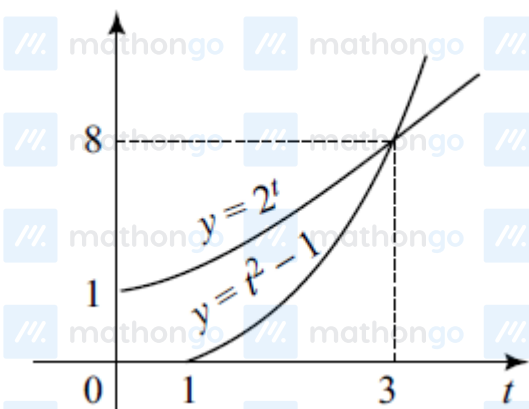
$$\Rightarrow (1 - \tan^4 \theta) + 2^{\tan^2 \theta} = 0$$

$$\Rightarrow 2^t = t^2 - 1 \text{ (where } t = \tan^2 \theta \text{)}$$

By inspection (or by graph) we find

$$y = 2^t \text{ and } y = t^2 - 1 \text{ intersect in } t = 3. \Rightarrow \tan^2 \theta = 3 \Rightarrow \tan \theta = \pm \sqrt{3}$$

$$\Rightarrow \theta = \pm \pi/3 \text{ (Fig.)}$$



Q23 (1,2,3,4) - Multiple Correct

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Let $81^{\sin^2 x} = y$.

Then $81^{\cos^2 x} = 81^{1-\sin^2 x}$

$= 81 \cdot y^{-1}$, so that the given equation is equivalent to $y + 81y^{-1} = 30$ or $y^2 - 30y + 81 = 0$

$\Rightarrow y = 3$ or $y = 27 \Rightarrow 81^{\sin^2 x} = 3$ or $81^{\sin^2 x} = 27$

$\Rightarrow 3^{4\sin^2 x} = 3$ or $3^{4\sin^2 x} = 3^3$

$\Rightarrow \sin^2 x = \frac{1}{4}$ or $\sin^2 x = \frac{3}{4}$

$\Rightarrow \sin x = \pm \frac{1}{2}$ or $\sin x = \pm \frac{\sqrt{3}}{2}$

which gives two solutions of the given equation in each quadrant.

Q24 (1,3,4) - Multiple Correct

As $-1 < \sin \theta < -\frac{\sqrt{3}}{2}$, $\tan(2\pi - \theta) > 0$, $\theta \in (0, 2\pi)$

We must have $\frac{3\pi}{2} < \theta < \frac{5\pi}{3}$. Now, $2 \cos \theta (1 - \sin \phi) = \sin^2 \theta \left(\tan \frac{\theta}{2} + \cot \frac{\theta}{2} \right) \cos \phi - 1$

$= 2 \sin^2 \theta \left[\frac{\tan^2(\theta/2) + 1}{2 \tan(\theta/2)} \right] \cos \phi - 1$

$= 2 \sin^2 \theta \frac{1}{\sin \theta} \cos \phi - 1$

$\Rightarrow 2 \cos \theta = 2[\sin \phi \cos \theta + \sin \theta \cos \phi] - 1$

$\Rightarrow 2 \cos \theta = 2 \sin(\phi + \theta) - 1$

As $\frac{3\pi}{2} < \theta < \frac{5\pi}{3}$, we get

$0 < \cos \theta < 1/2$

$\Rightarrow 0 < 2 \sin(\theta + \phi) - 1 < 1$

$\Rightarrow \frac{1}{2} < \sin(\theta + \phi) < 1$

As $\phi \in [0, 2\pi]$ and $\frac{3\pi}{2} < \theta < \frac{5\pi}{3}$,

$\theta + \phi \in \left(\frac{3\pi}{2}, \frac{11\pi}{3} \right)$

Thus, $\theta + \phi \in \left(\frac{13\pi}{6}, \frac{17\pi}{6} \right)$ [using (1)]

$\Rightarrow \frac{13\pi}{6} < \theta + \phi < \frac{17\pi}{6}$

Also, $\frac{5\pi}{3} < \theta < \frac{3\pi}{2}$

$\Rightarrow \frac{\pi}{2} < \phi < \frac{4}{3}\pi$

Therefore, ϕ cannot lie in $(0, \pi/2)$, $(4\pi/3, 3\pi/2)$ and $(3\pi/2, 2\pi)$.

Q25 (2,3) - Multiple Correct

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Adding the expression for y and z , we get

$$y + z = a (\cos^2 x + \sin^2 x) + c (\sin^2 x + \cos^2 x) \\ = a + c$$

and subtracting them $y - z = a (\cos^2 x - \sin^2 x) + 4b \sin x \cos x \\ - c (\cos^2 x - \sin^2 x)$

$$= a \cos 2x + 2b \sin 2x - c \cos 2x$$

$$= (a - c) \cos 2x + 2b \sin 2x$$

$$= (a - c) [\cos 2x + \tan x \sin 2x]$$

$$= (a - c) [\cos^2 x - \sin^2 x + 2 \sin^2 x]$$

$$= (a - c)$$

As $a \neq c$, we get $y \neq z$.

Q26 (1,2,3,4) - Multiple Correct

$$x = \sqrt{a^2 \cos^2 \alpha + b^2 \sin^2 \alpha} + \sqrt{a^2 \sin^2 \alpha + b^2 \cos^2 \alpha}$$

$$\Rightarrow x^2 = a^2 + b^2 + 2\sqrt{(a^2 \cos^2 \alpha + b^2 \sin^2 \alpha)(a^2 \sin^2 \alpha + b^2 \cos^2 \alpha)}$$

$$= a^2 + b^2 + k, \text{ where}$$

$$k = 2\sqrt{[(a^2 + b^2) - (a^2 \sin^2 \alpha + b^2 \cos^2 \alpha)] \times (a^2 \sin^2 \alpha + b^2 \cos^2 \alpha)}$$

$$\therefore x = a^2 + b^2 + 2\sqrt{(a^2 + b^2)P - P^2}$$

$$\text{where } P = a^2 \sin^2 \alpha + b^2 \cos^2 \alpha$$

$$\text{or } P = (a^2/2)(1 - \cos 2\alpha) + (b^2/2)(1 + \cos 2\alpha)$$

$$= (1/2)[a^2 + b^2 - (a^2 - b^2)\cos 2\alpha]$$

Also

$$x^2 = a^2 + b^2 + 2\sqrt{(a^2 \cos^2 \alpha + b^2 \sin^2 \alpha)[(a^2 + b^2) - (a^2 \cos^2 \alpha + b^2 \sin^2 \alpha)]}$$

$$= a^2 + b^2 + 2\sqrt{(a^2 + b^2)P - P^2}$$

where

$$P = a^2 \cos^2 \alpha + b^2 \sin^2 \alpha$$

$$\text{or } P = (1/2)[a^2 + b^2 + (a^2 - b^2)\cos 2\alpha]$$

Q27 (1,2,4) - Multiple Correct

The given expression can be written as

$$2[\cos \beta \cos \gamma + \cos \gamma \cos \alpha + \cos \alpha \cos \beta] +$$

$$2[\sin \beta \sin \gamma + \sin \gamma \sin \alpha + \sin \alpha \sin \beta] +$$

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$$\begin{aligned}
 &(\sin^2 \alpha + \cos^2 \alpha) + (\sin^2 \beta + \cos^2 \beta) + (\sin^2 \gamma + \cos^2 \gamma) = 0 \\
 \Rightarrow &(\cos \alpha + \cos \beta + \cos \gamma)^2 + (\sin \alpha + \sin \beta + \sin \gamma)^2 = 0 \\
 \Rightarrow &\Sigma \cos \alpha = 0 \text{ and } \Sigma \sin \alpha = 0 \\
 \Rightarrow &\Sigma(\cos \alpha + \sin \alpha) = 0
 \end{aligned}$$

Q28 (2) - Paragraph 1

$\sec \frac{\pi}{7}, \sec \frac{3\pi}{7}, \sec \frac{5\pi}{7}$ are the roots of the equation $x^3 - 4x^2 - 4x + 8 = 0$

$$\Rightarrow \sec \frac{\pi}{7} + \sec \frac{3\pi}{7} + \sec \frac{5\pi}{7} = 4$$

$$\text{and } \sec \frac{\pi}{7} \times \sec \frac{3\pi}{7} \times \sec \frac{5\pi}{7} = -8$$

$$\Rightarrow \sec \frac{\pi}{7} \sec \frac{3\pi}{7} \sec \frac{5\pi}{7} + 9 \sec \frac{\pi}{3} = -8 + 9(2) = 10$$

We can write

$$8x^3 - 4x^2 - 4x + 1 = 8 \left(x - \cos \frac{\pi}{7} \right) \left(x - \cos \frac{3\pi}{7} \right) \left(x - \cos \frac{5\pi}{7} \right)$$

Put $x = 1$

$$1 = 8 \left(2 \sin^2 \frac{\pi}{14} \times 2 \sin^2 \frac{3\pi}{14} \times 2 \sin^2 \frac{5\pi}{14} \right)$$

$$\Rightarrow \sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14} = \frac{1}{8}$$

Next, put $x = -1$, we get

$$\cos \frac{\pi}{14} \cos \frac{3\pi}{14} \cos \frac{5\pi}{14} = \frac{\sqrt{7}}{8}$$

Q29 (2) - Paragraph 1

$\sec \frac{\pi}{7}, \sec \frac{3\pi}{7}, \sec \frac{5\pi}{7}$ are the roots of the equation $x^3 - 4x^2 - 4x + 8 = 0$

$$\Rightarrow \sec \frac{\pi}{7} + \sec \frac{3\pi}{7} + \sec \frac{5\pi}{7} = 4$$

$$\text{and } \sec \frac{\pi}{7} \times \sec \frac{3\pi}{7} \times \sec \frac{5\pi}{7} = -8$$

$$\Rightarrow \sec \frac{\pi}{7} \sec \frac{3\pi}{7} \sec \frac{5\pi}{7} + 9 \sec \frac{\pi}{3} = -8 + 9(2) = 10$$

We can write

$$8x^3 - 4x^2 - 4x + 1 = 8 \left(x - \cos \frac{\pi}{7} \right) \left(x - \cos \frac{3\pi}{7} \right) \left(x - \cos \frac{5\pi}{7} \right)$$

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Hints & Solutions

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Put $x = 1$

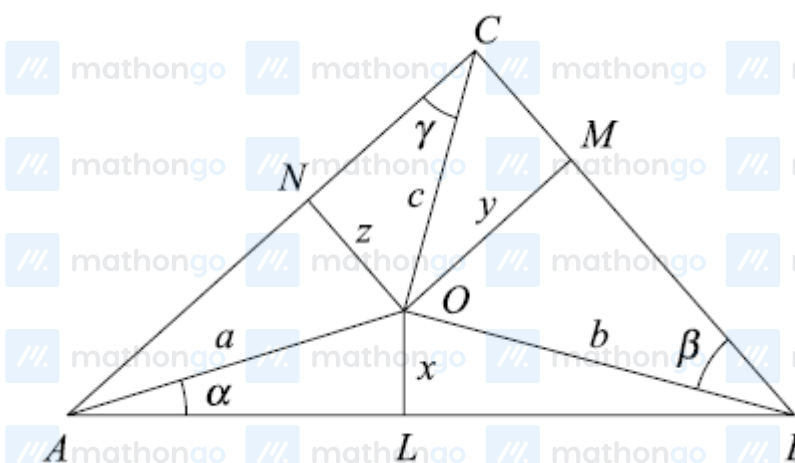
$$1 = 8 \left(2 \sin^2 \frac{\pi}{14} \times 2 \sin^2 \frac{3\pi}{14} \times 2 \sin^2 \frac{5\pi}{14} \right)$$

$$\Rightarrow \sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14} = \frac{1}{8}$$

Next, put $x = -1$, we get

$$\cos \frac{\pi}{14} \cos \frac{3\pi}{14} \cos \frac{5\pi}{14} = \frac{\sqrt{7}}{8}$$

Q30 (2) - Paragraph 2



$$AL = a \cos \alpha = x \cot \alpha$$

$$BM = b \cos \beta = y \cot \beta$$

$$CN = c \cos \gamma = z \cot \gamma$$

$$\sin(A - \alpha) = \frac{z}{a}, \sin(B - \beta) = \frac{x}{b}$$

$$\sin(C - \gamma) = \frac{y}{c}$$

$$a = x \operatorname{cosec} \alpha \text{ etc.}$$

$$\text{If } \alpha = A/3 \Rightarrow A = 3\alpha \Rightarrow \angle OAN = 2\alpha$$

$$\sin 2\alpha = \frac{z}{a}, \sin \alpha = \frac{x}{a}$$

$$\Rightarrow \frac{z}{a} = 2 \sin \alpha \cos \alpha = 2 \cdot \frac{x}{a} \sqrt{1 - \frac{x^2}{a^2}}$$

$$\Rightarrow az = 2x \sqrt{a^2 - x^2}$$

$$\sin A = \sin 3\alpha = 3 \sin \alpha - 4 \sin^3 \alpha = 3 \frac{x}{a} - 4 \frac{x^3}{a^3}$$

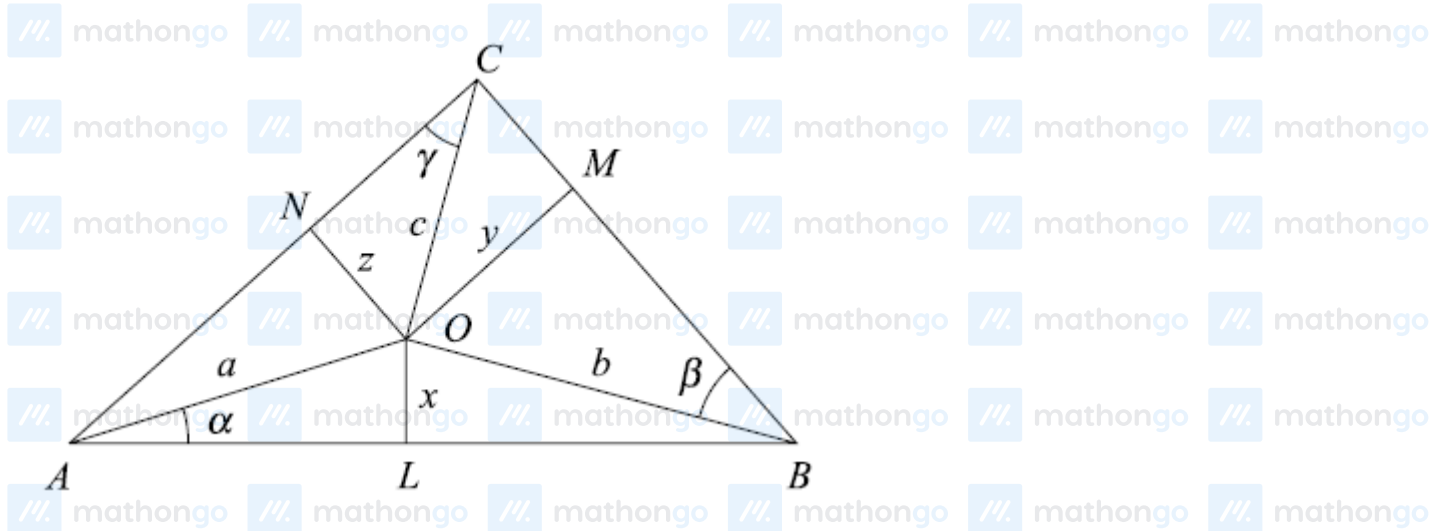
$$= \frac{3xa^2 - 4x^3}{a^3}$$

Q31 (3) - Paragraph 2

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$$AL = a \cos \alpha = x \cot \alpha$$

$$BM = b \cos \beta = y \cot \beta$$

$$CN = c \cos \gamma = z \cot \gamma$$

$$\sin(A - \alpha) = \frac{z}{a}, \sin(B - \beta) = \frac{x}{b}$$

$$\sin(C - \gamma) = \frac{y}{c}$$

$$a = x \operatorname{cosec} \alpha \text{ etc.}$$

$$\text{If } \alpha = A/3 \Rightarrow A = 3\alpha \Rightarrow \angle OAN = 2\alpha$$

$$\sin 2\alpha = \frac{z}{a}, \sin \alpha = \frac{x}{a}$$

$$\Rightarrow \frac{z}{a} = 2 \sin \alpha \cos \alpha = 2 \cdot \frac{x}{a} \sqrt{1 - \frac{x^2}{a^2}}$$

$$\Rightarrow az = 2x \sqrt{a^2 - x^2}$$

$$\sin A = \sin 3\alpha = 3 \sin \alpha - 4 \sin^3 \alpha = 3 \frac{x}{a} - 4 \frac{x^3}{a^3}$$

$$= \frac{3xa^2 - 4x^3}{a^3}.$$

Q32 (1) - Paragraph 3

$$f(x) = 0 \Rightarrow \sin\{\cot^{-1}(x+1)\} = \cos(\tan^{-1} x)$$

$$\Rightarrow \sin \sin^{-1} \frac{1}{\sqrt{1+(x+1)^2}} = \cos \cos^{-1} \frac{1}{\sqrt{1+x^2}}$$

$$\Rightarrow \frac{1}{\sqrt{1+(x+1)^2}} = \frac{1}{\sqrt{1+x^2}} \Rightarrow 1+x^2 = 2+x^2+2x$$

$$\Rightarrow x = -1/2, \text{ so } f(x) = 0 \text{ for } x = -1/2$$

$$a = \cos \tan^{-1} \sin \cot^{-1} x = \cos \tan^{-1} \sin \alpha \text{ where}$$

$$x = \cot \alpha.$$

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Hints & Solutions

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for $x = -1/2$

$$\text{Now } a^2 = 26/51 \Rightarrow x = \pm 1/5$$

$$\text{and } b = \cos(2 \cos^{-1} x + \sin^{-1} x) = \cos(\cos^{-1} x + \pi/2)$$

$$\Rightarrow b = -\sin(\cos^{-1} x) = -\sqrt{1-x^2}$$

$$\Rightarrow b^2 = 1 - x^2 = 24/25 \text{ for } x = \pm 1/5$$

Q33 (3) - Paragraph 3

$$f(x) = 0 \Rightarrow \sin\{\cot^{-1}(x+1)\} = \cos(\tan^{-1} x)$$

$$\Rightarrow \sin \sin^{-1} \frac{1}{\sqrt{1+(x+1)^2}} = \cos \cos^{-1} \frac{1}{\sqrt{1+x^2}}$$

$$\Rightarrow \frac{1}{\sqrt{1+(x+1)^2}} = \frac{1}{\sqrt{1+x^2}} \Rightarrow 1+x^2 = 2+x^2+2x$$

$$\Rightarrow x = -1/2, \text{ so } f(x) = 0 \text{ for } x = -1/2$$

$$a = \cos \tan^{-1} \sin \cot^{-1} x = \cos \tan^{-1} \sin \alpha \text{ where}$$

$$x = \cot \alpha.$$

for $x = -1/2$

$$\text{Now } a^2 = 26/51 \Rightarrow x = \pm 1/5$$

$$\text{and } b = \cos(2 \cos^{-1} x + \sin^{-1} x) = \cos(\cos^{-1} x + \pi/2)$$

$$\Rightarrow b = -\sin(\cos^{-1} x) = -\sqrt{1-x^2}$$

$$\Rightarrow b^2 = 1 - x^2 = 24/25 \text{ for } x = \pm 1/5$$

Q34 (1,3) - Paragraph 4

$$(2 \sin x - \cos x)(1 + \cos x) = \sin^2 x$$

$$\Rightarrow (1 + \cos x)[2 \sin x - \cos x - 1 + \cos x] = 0$$

$$\Rightarrow (1 + \cos x)(2 \sin x - 1) = 0$$

$$\Rightarrow \cos x = -1 \text{ or } \sin x = 1/2$$

$$\text{so } \sin \alpha = 1/2 \text{ [as } 0 \leq \alpha \leq \pi/2]$$

$$\Rightarrow \cos \alpha = \sqrt{3}/2$$

$$\text{Next, } 3 \cos^2 x - 10 \cos x + 3 = 0$$

$$\Rightarrow (3 \cos x - 1)(\cos x - 3) = 0$$

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$$\Rightarrow \cos x = 1/3 \text{ as } \cos x \neq 3$$

$$\text{So } \cos \beta = 1/3, \sin \beta = \frac{2\sqrt{2}}{3}$$

$$\text{and } 1 - \sin 2x = \cos x - \sin x$$

$$\Rightarrow \sin^2 x + \cos^2 x - 2 \sin x \cos x = \cos x - \sin x$$

$$\Rightarrow (\cos x - \sin x)(\cos x - \sin x - 1) = 0$$

$$\Rightarrow \text{Either } \sin x = \cos x \Rightarrow \sin \gamma = \cos \gamma = 1/\sqrt{2}$$

$$\text{or } \cos x - \sin x = 1 \Rightarrow \cos x = 1, \sin x = 0$$

$$\Rightarrow \cos \gamma = 1, \sin \gamma = 0$$

So that $\cos \alpha + \cos \beta + \cos \gamma$ can be equal to

$$\frac{\sqrt{3}}{2} + \frac{1}{3} + \frac{1}{\sqrt{2}} \text{ or } \frac{\sqrt{3}}{2} + \frac{1}{3} + 1$$

$$\text{i.e. } \frac{3\sqrt{6}+2\sqrt{2}+6}{6\sqrt{2}} \text{ or } \frac{3\sqrt{3}+8}{6}$$

$\sin \alpha + \sin \beta + \sin \gamma$ can be equal to

$$\frac{1}{2} + \frac{2\sqrt{2}}{3} + \frac{1}{\sqrt{2}} \text{ or } \frac{1}{2} + \frac{2\sqrt{2}}{3} + 0$$

$$\text{i.e. } \frac{3\sqrt{2}+14}{6\sqrt{2}} \text{ or } \frac{3+4\sqrt{2}}{6}$$

and $\sin(\alpha - \beta)$ is equal to $\sin \alpha \cos \beta - \cos \alpha \sin \beta$

$$= \frac{1}{2} \times \frac{1}{3} - \frac{\sqrt{3}}{2} \times \frac{2\sqrt{2}}{3} = \frac{1-2\sqrt{6}}{6}.$$

Q35 (3) - Paragraph 4

$$(2 \sin x - \cos x)(1 + \cos x) = \sin^2 x$$

$$\Rightarrow (1 + \cos x)[2 \sin x - \cos x - 1 + \cos x] = 0$$

$$\Rightarrow (1 + \cos x)(2 \sin x - 1) = 0$$

$$\Rightarrow \cos x = -1 \text{ or } \sin x = 1/2$$

$$\text{so, } \sin \alpha = 1/2 \text{ [as } 0 \leq \alpha \leq \pi/2]$$

$$\Rightarrow \cos \alpha = \sqrt{3}/2$$

$$\text{Next, } 3 \cos^2 x - 10 \cos x + 3 = 0$$

$$\Rightarrow (3 \cos x - 1)(\cos x - 3) = 0$$

$$\Rightarrow \cos x = 1/3 \text{ as } \cos x \neq 3$$

$$\text{So } \cos \beta = 1/3, \sin \beta = \frac{2\sqrt{2}}{3}$$

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$$\text{and } 1 - \sin 2x = \cos x - \sin x$$

$$\Rightarrow \sin^2 x + \cos^2 x - 2 \sin x \cos x = \cos x - \sin x$$

$$\Rightarrow (\cos x - \sin x)(\cos x - \sin x - 1) = 0$$

$$\Rightarrow \text{Either } \sin x = \cos x \Rightarrow \sin \gamma = \cos \gamma = 1/\sqrt{2}$$

$$\text{or } \cos x - \sin x = 1 \Rightarrow \cos x = 1, \sin x = 0$$

$$\Rightarrow \cos \gamma = 1, \sin \gamma = 0$$

So that $\cos \alpha + \cos \beta + \cos \gamma$ can be equal to

$$\frac{\sqrt{3}}{2} + \frac{1}{3} + \frac{1}{\sqrt{2}} \text{ or } \frac{\sqrt{3}}{2} + \frac{1}{3} + 1$$

$$\text{i.e. } \frac{3\sqrt{6}+2\sqrt{2}+6}{6\sqrt{2}} \text{ or } \frac{3\sqrt{3}+8}{6}$$

$\sin \alpha + \sin \beta + \sin \gamma$ can be equal to

$$\frac{1}{2} + \frac{2\sqrt{2}}{3} + \frac{1}{\sqrt{2}} \text{ or } \frac{1}{2} + \frac{2\sqrt{2}}{3} + 0$$

$$\text{i.e. } \frac{3\sqrt{2}+14}{6\sqrt{2}} \text{ or } \frac{3+4\sqrt{2}}{6}$$

and $\sin(\alpha - \beta)$ is equal to $\sin \alpha \cos \beta - \cos \alpha \sin \beta$

$$= \frac{1}{2} \times \frac{1}{3} - \frac{\sqrt{3}}{2} \times \frac{2\sqrt{2}}{3} = \frac{1-2\sqrt{6}}{6}.$$

Q36 (3) - Paragraph 5

$$\text{We have } 2 \cos 2\theta(\cos 4\theta + \cos 2\theta) = 1$$

$$\Rightarrow 2 \cos 2\theta \cos 4\theta + 2 \cos^2 2\theta - 1 = 0$$

$$\Rightarrow \cos 4\theta(2 \cos 2\theta + 1) = 0$$

$$\Rightarrow \text{Either } \cos 4\theta = 0 \text{ or } \cos 2\theta = -1/2$$

$$\cos 4\theta = 0 \Rightarrow \theta = (2n+1)\pi/8n \in \mathbf{I}$$

$$\Rightarrow \theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8} \text{ as } 0 \leq \theta \leq \pi \text{ and}$$

$$\cos 2\theta = -1/2$$

$$\Rightarrow \theta = \pi/3 \text{ and } 2\pi/3 \text{ again as } 0 \leq \theta \leq \pi$$

So the sum of the roots = 3π

Next, Let $x = 2 \cos \alpha$ where α is a root of the equation i.e. either $\cos 4\alpha = 0$ or $\cos 2\alpha = -1/2$

$$\text{so } x^2 = 4 \cos^2 \alpha = 2(1 + \cos 2\alpha) = 1 \text{ if } \cos 2\alpha = -1/2$$

$$\Rightarrow 2 \cos \alpha \text{ is a root of the equation } x^2 - 1 = 0$$

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or $x^2 - 2 = 2 \cos 2\alpha$.

$$\Rightarrow (x^2 - 2)^2 = 4 \cos^2 2\alpha = 2(1 + \cos 4\alpha) = 2 \text{ if } \cos 4\alpha = 0$$

$$\Rightarrow x^4 - 4x^2 + 2 = 0 \Rightarrow 2 \cos \alpha \text{ is a root of the equation}$$

$$x^4 - 4x^2 + 2 = 0$$

$$\text{Now } 3 \cos 2\theta = 1 \Rightarrow \cos 2\theta = 1/3$$

$$\Rightarrow \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{1}{3} \Rightarrow \tan^2 \theta = \frac{1}{2}$$

So from $32 \tan^8 \theta = 2 \cos^2 \alpha - 3 \cos \alpha$, we get

$$2 = 2 \cos^2 \alpha - 3 \cos \alpha \Rightarrow \cos \alpha = -1/2 \text{ as } \cos \alpha \neq 2$$

which is satisfied by $\alpha = 2\pi/3$, a root of the given equation.

Q37 (1,3) - Paragraph 5

$$\text{We have } 2 \cos 2\theta (\cos 4\theta + \cos 2\theta) = 1$$

$$\Rightarrow 2 \cos 2\theta \cos 4\theta + 2 \cos^2 2\theta - 1 = 0$$

$$\Rightarrow \cos 4\theta (2 \cos 2\theta + 1) = 0$$

$$\Rightarrow \text{Either } \cos 4\theta = 0 \text{ or } \cos 2\theta = -1/2$$

$$\cos 4\theta = 0 \Rightarrow \theta = (2n + 1)\pi/8 \in \mathbf{I}$$

$$\Rightarrow \theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8} \text{ as } 0 \leq \theta \leq \pi \text{ and}$$

$$\cos 2\theta = -1/2$$

$$\Rightarrow \theta = \pi/3 \text{ and } 2\pi/3 \text{ again as } 0 \leq \theta \leq \pi$$

So the sum of the roots = 3π

Next, Let $x = 2 \cos \alpha$ where α is a root of the equation i.e. either $\cos 4\alpha = 0$ or $\cos 2\alpha = -1/2$

$$\text{so } x^2 = 4 \cos^2 \alpha = 2(1 + \cos 2\alpha) = 1 \text{ if } \cos 2\alpha = -1/2$$

$$\Rightarrow 2 \cos \alpha \text{ is a root of the equation } x^2 - 1 = 0$$

or $x^2 - 2 = 2 \cos 2\alpha$.

$$\Rightarrow (x^2 - 2)^2 = 4 \cos^2 2\alpha = 2(1 + \cos 4\alpha) = 2 \text{ if } \cos 4\alpha = 0$$

$$\Rightarrow x^4 - 4x^2 + 2 = 0 \Rightarrow 2 \cos \alpha \text{ is a root of the equation}$$

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$$x^4 - 4x^2 + 2 = 0$$

$$\text{Now } 3 \cos 2\theta = 1 \Rightarrow \cos 2\theta = 1/3$$

$$\Rightarrow \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{1}{3} \Rightarrow \tan^2 \theta = \frac{1}{2}$$

So from $32 \tan^8 \theta = 2 \cos^2 \alpha - 3 \cos \alpha$, we get

$$2 = 2 \cos^2 \alpha - 3 \cos \alpha \Rightarrow \cos \alpha = -1/2 \text{ as } \cos \alpha \neq 2$$

which is satisfied by $\alpha = 2\pi/3$, a root of the given equation.

Q38 (8) - Integer Type

$$\cos^4 x + \sin^4 x$$

$$= (\cos^2 x + \sin^2 x)^2 - 2 \cos^2 x \sin^2 x$$

$$= 1 - \frac{1}{2} \sin^2 2x$$

$$\text{and } \cos^6 x + \sin^6 x$$

$$= (\cos^2 x + \sin^2 x)^3 - 3 \cos^2 x \sin^2 x (\cos^2 x + \sin^2 x)$$

$$= 1 - \frac{3}{4} \sin^2 2x$$

Thus, the given equation becomes

$$\frac{5}{4} \cos^2 2x + 1 - \frac{1}{2} \sin^2 2x + 1 - \frac{3}{4} \sin^2 2x = 2$$

$$\Rightarrow \sin^2 2x = \cos^2 2x$$

$$\Rightarrow \tan(2x) = \pm 1$$

$$\Rightarrow 2x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}, \frac{11\pi}{4}, \frac{13\pi}{4}, \frac{15\pi}{4}$$

$$\Rightarrow x = \frac{1}{8}(2k-1)\pi \text{ where } k = 1, 2, 3, 4, 5, 6, 7, 8.$$

Thus, number of distinct solution is 8.

Q39 (3) - Integer Type

$$\tan \theta = \cot 5\theta$$

$$\Rightarrow \frac{\sin \theta}{\cos \theta} = \frac{\cos 5\theta}{\sin 5\theta}$$

$$\Rightarrow 2 \cos 5\theta \cos \theta = 2 \sin 5\theta \sin \theta$$

$$\Rightarrow \cos 6\theta + \cos 4\theta = \cos 4\theta - \cos 6\theta$$

$$\Rightarrow 2 \cos 6\theta = 0 \Rightarrow \cos 6\theta = 0$$

$$\text{As } -\frac{\pi}{2} < \theta < \frac{\pi}{2} \text{ we get } -3\pi < 6\theta < 3\pi$$

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$$\text{Now, } \cos 6\theta = 0 \Rightarrow 6\theta = \pm \frac{5\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{\pi}{2}$$

$$\Rightarrow \theta = \pm \frac{5\pi}{12}, \pm \frac{\pi}{4}, \pm \frac{\pi}{12}$$

But $\theta = -\frac{5\pi}{12}, -\frac{\pi}{12}$ and $\frac{\pi}{4}$ do not satisfy the relation $\sin 2\theta = \cos 4\theta$

Note that

$$\theta = \frac{5\pi}{12}, -\frac{\pi}{4} \text{ and } \frac{\pi}{12}$$

Satisfy the relation $\sin 2\theta = \cos 4\theta$.

Q40 (3) - Integer Type

The three equations can be written as

$$(y+z) \cos 3\theta = xyz \sin 3\theta$$

$$2y \sin 3\theta + 2z \cos 3\theta = xyz \sin 3\theta$$

$$(y+2z) \cos 3\theta + y \sin 3\theta = xyz \sin 3\theta$$

$$\text{Thus, } (y+z) \cos 3\theta = 2y \sin 3\theta + 2z \cos 3\theta$$

$$y(2 \sin 3\theta - \cos 3\theta) + z \cos 3\theta = 0$$

$$\Rightarrow y(\cos 3\theta - \sin 3\theta) = 0$$

As $y_0 z_0 \neq 0$, we take $y = y_0 \neq 0$, so that $\cos 3\theta = \sin 3\theta$

$$\Rightarrow \tan 3\theta = 1 \Rightarrow 3\theta = n\pi + \frac{\pi}{4}$$

$$\Rightarrow \theta = \frac{1}{3}n\pi + \frac{\pi}{12}$$

As $0 < \theta < \pi$, we get, three value of θ .

Q41 (3) - Integer Type

We have $f(x) = \cos^{-1}(\cos x)$

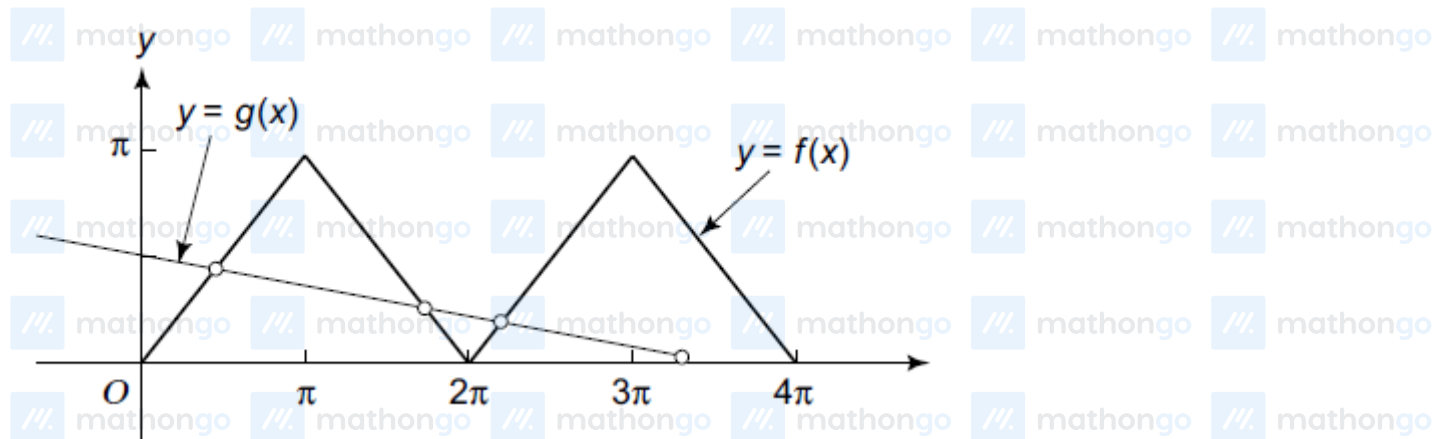
$$= \begin{cases} \cos^{-1}(\cos x) & \text{if } 0 \leq x \leq \pi \\ \cos^{-1}(\cos(2\pi - x)) & \text{if } \pi \leq x \leq 2\pi \\ \cos^{-1}(\cos(x - 2\pi)) & \text{if } 2\pi \leq x \leq 3\pi \\ \cos^{-1}(\cos(4\pi - x)) & \text{if } 3\pi \leq x \leq 4\pi \end{cases}$$

$$= \begin{cases} x & \text{if } 0 \leq x \leq \pi \\ 2\pi - x & \text{if } \pi \leq x \leq 2\pi \\ x - 2\pi & \text{if } 2\pi \leq x \leq 3\pi \\ 4\pi - x & \text{if } 3\pi \leq x \leq 4\pi \end{cases}$$

Graph of $y = f(x)$ and $y = \frac{10-x}{10} = 1 - \frac{x}{10} = g(x)$ are shown in the following figure.

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The above figure shows that the equation $f(x) = \frac{10-x}{x}$ has 3 solutions.

Q42 (5) - Integer Type

$$x = 1 + \tan^2(\tan^{-1} 2) + 1 + \cot^2(\cot^{-1} 3)$$

$$= 1 + 4 + 1 + 9 = 15$$

$$y = \tan^{-1} \frac{1 - \cos 8}{\sin 8} = \tan^{-1} \tan 4 = 4$$

$$\therefore 15x^2 + 4y^2 = (15)^3 + (4)^3 = 3439 = 3434 + \lambda \Rightarrow \lambda = 5$$

Q43 (0) - Integer Type

$$\cos \theta (1 + \cos^2 \theta) = \sin^2 \theta$$

$$\Rightarrow (1 - \sin^2 \theta) (2 - \sin^2 \theta)^2 = \sin^4 \theta$$

$$\Rightarrow (1 - \sin^2 \theta) (4 - 4 \sin^2 \theta + \sin^4 \theta) = \sin^4 \theta$$

$$\Rightarrow \sin^6 \theta = 4 - 8 \sin^2 \theta + 4 \sin^4 \theta$$

Comparing with the given equation we get

$$a = 4, b = -8, c = 4$$

$$\Rightarrow a + b + c = 0$$

Q44 (3) - Integer Type

$$K = \frac{1}{4} [(3 \sin 10^\circ - \sin 30^\circ)$$

$$+ (3 \sin 50^\circ - \sin 150^\circ) - (3 \sin 70^\circ - \sin 210^\circ)]$$

$$= \frac{1}{4} \left[3 (\sin 10^\circ + \sin 50^\circ - \sin 70^\circ) - \frac{3}{2} \right]$$

$$= \frac{1}{4} \left[3 (\sin 10^\circ - 2 \cos 60^\circ \sin 10^\circ) - \frac{3}{2} \right]$$

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$$= -\frac{3}{8}$$

$$\Rightarrow 8K + 6 = 3$$

Q45 (1) - Integer Type

From conditional identities we have

$$\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = \frac{4 \sin A \sin B \sin C}{4 \cos(A/2) \cos(B/2) \cos(C/2)}$$

$$= 8 \sin(A/2) \sin(B/2) \sin(C/2)$$

$$\Rightarrow k = 8$$

$$\text{and } k^3 - 8k^2 + 1 = 1$$